Khadijah Aamir Chaudhry

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EDUCATION

Lahore University of Management Sciences (LUMS) *2023–2027*
BSc Computer Science · Minor in Computer Engineering

Lahore Grammar School *2018–2023*
A Levels: 4 As · O Levels: 9 A*s · SAT: 1530

RESEARCH INTERESTS

Secure embedded systems · Hardware roots of trust · Hardware–software co-design · Trusted execution · Applied cryptography

RESEARCH EXPERIENCE

Research Assistant, CHASE Lab, LUMS *Aug 2025–present*
Supervisor: Dr. Muhammad Ali Siddiqi

- Benchmarked **post-quantum algorithms on ESP32** to compare latency, memory, network cost, and estimated energy on constrained devices.
- Built a **simulated RISC-V trace testbed** to capture transaction-level activity for research into runtime hardware Trojan detection.

Research Assistant, SysNet Lab, LUMS *Jan–May 2026*
Supervisor: Dr. Naveed Anwar Bhatti

- Co-developed **EyeSense's smartphone ToF/ML pipeline** for refractive-error screening to explore more accessible eyesight measurement.
- Helped collect **eyesight data in lab and eye-camp settings**, supporting evaluation of phone-based screening with real measurements.

SELECTED PROJECTS

On Chip Behavioral Monitoring *2026–present*

- Built a Rocket Chip/Verilator testbed with four RV32 configurations and **TileLink MMIO trace capture** for runtime hardware monitoring research.
- Replayed **benign and trigger-active firmware** to collect contrasting traces for later detector evaluation.

Protected Media Containers *2026–present*

- Developed a system design for **protected camera/audio capture**, encrypted storage, display, and sharing under an untrusted mobile OS.
- Reviewed **trusted-execution and DRM research** to define how privacy controls can persist as media is stored and shared.

EyeSense: Mobile Eyesight Screening *Jan–May 2026*

- Co-developed a **smartphone ToF phase-map and machine-learning pipeline** for refractive-error screening, exploring a phone-based approach to eyesight checks.
- Supported a team study of 120 participants that reported **1.15 D mean absolute error**, providing an initial measure of screening performance.

Post-Quantum Cryptography on ESP32 *Aug–Dec 2025*

- Benchmarked **Kyber-512, Light-SABER, and Dilithium-II** on ESP32 to assess post-quantum cryptography on resource-constrained hardware.
- Compared **latency, memory, network cost, and estimated energy use** to document practical trade-offs for small connected devices.

AasPass Peer Tutoring

- Led authentication and administration features for an Android student–tutor app, including **LUMS email verification and profile setup**.
- Added **moderation controls** so administrators could manage accounts and support the peer tutoring community.

Spotify Mood Classification and Recommendation *Jan–May 2025*

- Applied **PCA and K-means** to audio features to explore how songs group by mood and musical similarity.
- Built **mood classification and similarity-based recommendations** to help listeners discover songs with a similar feel.

ROM-Based Digital Lock

- Built and tested an **EPROM-based keypad lock** using state-machine logic to control access based on an entered code.
- Added an **unlock indicator and an incorrect-input alarm** to make accepted and rejected codes clear to users.

TEACHING EXPERIENCE

Teaching Assistant, CS100, LUMS *Aug 2026–present*

- Designed and conducted programming labs and prepared and graded quizzes to support about **450 first-year students** learning introductory programming.
- Held **weekly office hours** and invigilated exams, providing regular opportunities for students to seek help and receive assessment support.

PROFESSIONAL EXPERIENCE

Marketing Manager, Fintelligent Solutions *Jul 2026–present*

- Coordinated team delivery of **brand identity and the company website** for a Microsoft Dynamics 365 consultancy, helping clients understand its services.
- Coordinated **office collateral and LinkedIn launch content** to give the consultancy consistent materials for client outreach.

Data Collection Coach, Uplift AI *Jul–Aug 2024*

- Led **10 data collectors** and organized field outreach for a Pakistani-language AI dataset, helping gather locally relevant voice data.
- Ran **daily quality reviews and feedback sessions** to identify collection issues and improve consistency in submitted recordings.

LEADERSHIP & SERVICE

Peer Ambassador Leader, LUMS *Aug 2026–present*

- Guided **15 incoming students** through orientation and faculty introductions to help them settle into university life.
- Conduct **monthly first-year support sessions** to give students a familiar point of contact as they navigate university.

Women in STEM Coordinator, LUMS SPADES *Sep 2024–May 2025*

- Helped launch the **first women-only STEM event at PSIFI** for more than 100 participants, creating a dedicated space for women to engage with STEM.
- Led the Women in STEM department in organizing bonding events and **digital and financial literacy seminars** to support participation and practical learning.

Co-Founder and Logistics Director, Umeed-e-Nau *Sep 2021–May 2023*

- Co-founded Umeed-e-Nau, an NGO providing **inclusive educational activities** for underprivileged and disabled children.
- Directed logistics for **tailored learning activities and visits**, including accommodations that helped disabled children participate.

SELECTED GRADUATE COURSEWORK

Internet of Things · Applied Cryptography · Topics in Computer and Network Security · Distributed Systems

HONORS AND AWARDS

- **100% Merit Scholarship**, Lahore Grammar School *Aug 2021–May 2023*
- **Shortlisted writing submission**, LUMSx Nexus *May 2025*
- **Honorable Mention**, EducationUSA Pakistan Writing Contest *Sep 2022*

TECHNICAL SKILLS

Programming languages: C · C++ · Python · JavaScript
Embedded and hardware: ESP32 · RISC-V · Rocket Chip · Verilator · FreeRTOS
Machine learning and sensing: ToF sensing · CNNs · ridge regression
Tools and methods: Docker · Git · post-quantum cryptography benchmarking